

Primates

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Primates
FAMILIES	12
SPECIES	480

CLASSIFICATION NOTE

Recent research suggests that primates may comprise 2 suborders: Strepsirrhini (lemurs, galagos, lorises, and pottos) and Haplorrhini (tarsiers, apes, and monkeys). Tarsiers have features linking them with both suborders; in this book, they are grouped with Strepsirrhini (traditionally known as the prosimians). To allow adequate coverage, monkeys and apes are presented here as separate groups. Humans do not appear in this book but are classified, with the apes, as hominoids.

Prosimians see pp. 136–9
Monkeys see pp. 140–9
Apes see pp. 150–5

Members of the order Primates—prosimians, monkeys, and apes (but see the Classification note, left)—are a diverse group that form highly complex social units. They are found in South and Central America, Africa and Madagascar, and Southeast and eastern Asia. Primates mostly inhabit tropical rain forests, and their dextrous prehensile (grasping) hands and feet are an adaptation to a largely arboreal lifestyle (some also have a prehensile tail). Over the last 25 years, 96 new species of primate have been described, nearly half of them being found in Madagascar—an area where many species are in danger of extinction due to increasing human activity. Deforestation here and elsewhere in the tropics is severe, reducing and degrading primate habitat to the extent that it is predicted that 60 percent of all primate species could disappear over the next 25–50 years.

Social organization

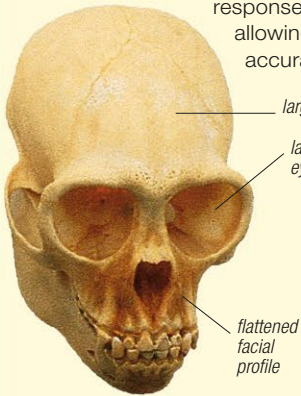
Only orangutans and a few of the lemurs and galagos lead solitary lives; all other primates exist in social groups. Many species—including most monkeys—live in troops that consist of several females and either one or several adult males. In some species, such as the mandrill, drill, and gelada, huge troops of several hundred individuals periodically split into subgroups of bachelor bands and one male and his harem, both of fixed composition. Chimpanzees and spider monkeys live in large communities of 20–100 individuals that divide into groups of varying composition. Some species, mostly New World monkeys, live in monogamous pairs.



MUTUAL GROOMING
The bonds between members of primate groups are partly maintained by mutual grooming, as shown by these olive baboons. Low-ranking members groom those of higher rank to gain favor and support (in disputes).

Anatomy

Primates form a highly varied group, with members as diverse as mouse lemurs, which may weigh only 1¼ oz (35 g), and gorillas, which can reach over 440 lb (200 kg). Most species have flat nails on the fingers and toes—only a few have claws—and the big toe always has a nail. Most have a tail, except the apes. Apart from some toothed whale species, the higher primates have the largest brain relative to body size of all the mammals, which helps explain their high intelligence. The cerebral hemispheres (which process sensory information and coordinate responses) are highly developed, allowing sight keen enough for accurate tree-to-tree leaps.



large braincase
large, forward-facing eye sockets
flattened facial profile

SKULL STRUCTURE

Primates have a large, domed braincase and forward-facing eye sockets. This monkey skull also shows the flattened facial profile that is characteristic of most monkeys and apes.

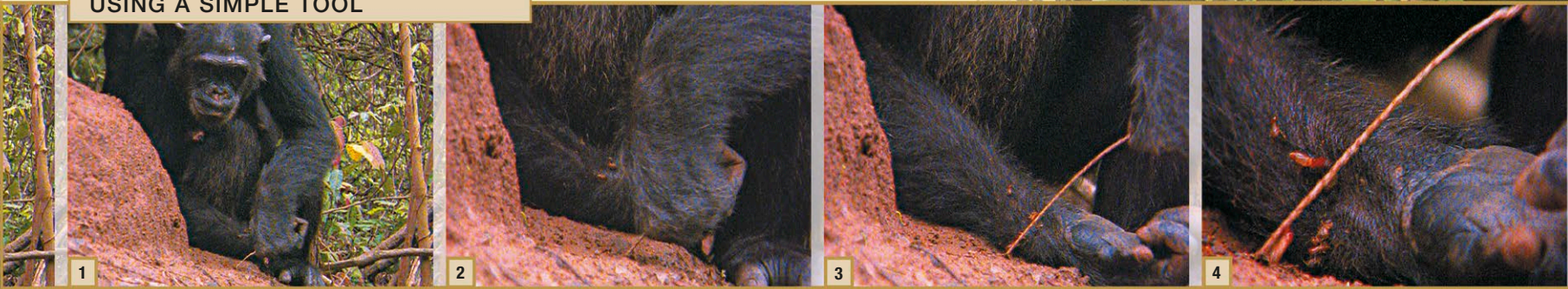


HANDS AND FEET

Hand and foot structure varies according to lifestyle. Claws (aye-aye) and rounded pads (tarsier) improve grip in arboreal species; highly mobile hands and feet (chimpanzee) are vital for species that live on the ground and in trees.



USING A SIMPLE TOOL



SEEKING TERMITES

This common chimpanzee has made a stick tool to obtain food that would otherwise be inaccessible.

DEXTRIOUS HANDS

The chimpanzee uses its grasping hands to hold and insert the stick into the termite mound.

REMOVING THE TOOL

The termites on guard, provoked by the intrusion, bite the stick with their pincerlike jaws.

SUCCESSFUL ENDEAVOR

As the chimpanzee removes the stick, the termites remain firmly attached.